

AASHISH DHAWAN

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SUMMARY

PhD researcher (UF Data Science & Research Lab, Advisor: Dr. Daisy Wang) working on multimodal and foundation models, retrieval-augmented learning, AI reasoning, optimization, and trustworthy AI. **AmericasNLP 2026 shared task winner** (ACL) and **top-performing system at WMT26 Indic MT** (EMNLP). Published across ACL, EACL, and EMNLP workshops; WMT26 paper forthcoming at EMNLP 2026. Built reusable production XAI assets at ExxonMobil for enterprise-scale demand forecasting and mathematical optimization.

Available: Summer 2027 internships (May–Aug 2027); full-time from May 2028. F-1, CPT/OPT eligible; no sponsorship required for internships, H-1B sponsorship required for full-time.

EDUCATION

Ph.D. in Computer Science

University of Florida · Data Science & Research Lab

Aug 2025 – May 2028 (Expected)

Gainesville, FL

• Advisor: Dr. Daisy Wang Research: Multimodal AI, Low-Resource NLP, Retrieval-Augmented Generation

M.S. in Computer Science

University of Florida

Jan 2021 – May 2023

Gainesville, FL

B.Tech. in Computer Science

Kurukshetra University

Aug 2015 – May 2019

Haryana, India

PUBLICATIONS

- **Dhawan et al.** “BM25-Augmented Many-Shot Translation for Low-Resource North-Eastern Indian Languages.” *WMT26 Indic MT Shared Task @ EMNLP 2026, Budapest*. [Top-Performing System]
- **Dhawan et al.** “Retrieval-Augmented Long-Context Translation for Cultural Image Captioning.” *AmericasNLP @ ACL 2026, San Diego*. ACL Anthology. [1st Place, Overall Shared Task]
- **Dhawan et al.** “Improving Indigenous Language Machine Translation with Synthetic Data and Language-Specific Preprocessing.” *LoResMT @ EACL 2026*. ACL Anthology.
- Driggers-Ellis, . . . , **Dhawan et al.** “MultiScript30k: Multilingual Embeddings for Cross-Script Parallel Data.” arXiv:2512.11074. *Under review, ACL ARR*.
- **Dhawan et al.** “Post Processing of Image Segmentation Using CRF.” *IEEE INDIACom*, 2019.

EXPERIENCE

Computational Data Scientist Intern

ExxonMobil

May 2026 – Aug 2026

Houston, TX

- **Beat the incumbent demand forecaster by 5.4 pp MAPE** (20.9% → 15.4%, 970 held-out samples) and shipped it as a reusable production explainability asset: LightGBM with SHAP, LIME, counterfactual, and drift-detection analysis translating predictions into verified plain-English narratives for operations teams that did not trust AI recommendations.
- Built an explainability framework from scratch for the LP and MILP optimization behind the same demand forecasting business (Gurobi, Pyomo/IPOPT): which constraints bind, why an option was not chosen, and how outputs shift when a limit is relaxed, with a verification layer that surfaces only claims it can prove from solver output.
- Red-teamed both tools with adversarial inputs to confirm explanations cannot be coerced into fabricating numbers; built on a plug-in model interface so they extend to other product lines and teams.

Machine Learning Researcher

Data Science & Research Lab, University of Florida

May 2024 – Present

Gainesville, FL

- Built a generalizable **BM25 + many-shot RAG translation pipeline** entered in **two international shared tasks on the same core architecture**: AmericasNLP 2026 (Qwen2.5-VL multimodal captioning, ACL 2026, **1st place overall**) and WMT26 Indic MT (text-only, EMNLP 2026, **top-performing system**), showing that retrieval-augmented few-shot prompting transfers across low-resource language families and modalities.
- Fine-tuned multilingual **mBART-50** with distributed data-parallel (DDP) multi-GPU training, cutting iteration time enough to run a **full ablation study** with chrF++ evaluation, published at EACL 2026.
- Engineered a few-shot object detection system for low-resource visual domains (CLIP + SAM + DINO + Adaptive Feature Transfer + OHAC clustering), achieving **15% accuracy gain** over CLIP zero-shot with concept-level visual explanations per prediction.

Adjunct Instructor

University of Florida

Aug 2023 – Present

Gainesville, FL

- Designed and delivered Machine Learning and Programming Languages to **300+ students/semester**; led 5+ TAs; mentored capstone projects in LLM chatbots, real-time computer vision, and recommendation systems.

AI Engineer / Consultant

Aug 2023 – Apr 2024

Atmosphere Apps

Gainesville, FL

- Engineered GPT-4 and Whisper NLP pipelines for multilingual transcription and prescription recommendation; **95% accuracy** across 10+ languages for **5,000+ active users**.

Graduate Research Assistant

Jan 2021 – May 2023

Graphics Imaging & Light Measurement Lab, University of Florida

Gainesville, FL

- Deep learning pipelines for multispectral image synthesis (ProGAN), neural quantum simulation (**82% compute reduction**), and material scattering prediction via differentiable rendering.

Visiting Researcher

Dec 2019 – Mar 2020

UBTECH AI Lab, University of Sydney

Sydney, Australia

- Multi-source domain adaptation (CNN + GRL adversarial training + moment matching); 71–98% accuracy across 345 classes on a 600k-sample benchmark. Advisor: Prof. Dacheng Tao.

Research Intern

May 2019 – Aug 2019

Space Applications Center, ISRO

Ahmedabad, India

- CNN-CRF satellite segmentation pipeline (LISS-3 + POTSDAM): replaced a **36-hour CRF post-processing stage with a 0.2s learned approximation** while improving segmentation accuracy by **5%**. Published IEEE INDIACom 2019.

SKILLS

Languages: Python, SQL, Java, C++, R

LLMs & Generative AI: Fine-tuning (LoRA/PEFT), RAG, Many-Shot / Few-Shot Prompting, LangChain

Multimodal & Vision: Diffusion Models, Vision-Language Models, Object Detection, Image Segmentation

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LightGBM, XGBoost, Distributed Training (DDP, multi-GPU)

Trustworthy AI & XAI: SHAP, LIME, Counterfactual Explanations, Verification, Adversarial Red-Teaming, Drift Detection

Optimization: Gurobi, Pyomo/IPOPT, Linear and Mixed-Integer Programming

Tools & MLOps: FastAPI, MCP, AWS, Docker, Kubernetes, Git, Streamlit, Pandas, NumPy

SELECTED PROJECTS

Visual Multi-Step Reasoning with MLM-U | *PyTorch, Transformers, CNNs, Hydra*

2026

- Extended Meta's PAST/MLM-U architecture from text-only mazes to ViT-derived visual tokens while retaining text-based path prediction; benchmarked autoregressive and multi-token prediction across maze scales to isolate perception, planning, generalization, and convergence bottlenecks in visual reasoning.

Explainable Few-Shot Detection for Low-Resource Domains | *PyTorch, CLIP, SAM, DINO*

2025

- Concept-driven detector for indigenous manuscript glyphs: CLIP + SAM + DINO + Adaptive Feature Transfer + OHAC clustering; **15% accuracy gain** over CLIP zero-shot with hierarchical concept-level explanations served via FastAPI.

AWARDS & RECOGNITION

- 1st Place, AmericasNLP 2026 Shared Task** @ ACL 2026, San Diego: best overall system across all competing teams.
- Top-performing system, WMT26 Indic MT Shared Task** @ EMNLP 2026, Budapest.